Python-Numpy

Objective

- NumPy Arrays
- NumPy Arrays v/s List
- NumPy Attributes
- Array Slicing/Indexing
- NumPy Operations
- NumPy Special Functions
- Summary

NumPy Arrays

- NumPy stands for numeric python which is a python package for the computation and processing of the multidimensional and single dimensional array elements.
- NumPy's main object is the homogeneous multidimensional array.
- Numpy array is a powerful N-dimensional array object which is in the form of rows and columns.
- It is a table of elements (usually **numbers**), all of the same type, indexed by a tuple of positive integers.
- In NumPy dimensions are called **axes**. The number of axes is **rank**.
- NumPy's array class is called **ndarray**. It is also known by the alias **array**.

- Example :

  [[1, 2, 3],

  [4, 2, 5]]

  Here,

- rank = 2 (as it is 2-dimensional or it has 2 axes)
- first dimension(axis) length = 2, second dimension has length = 3
- overall shape can be expressed as: (2, 3)

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Installation NumPy

- To install Python NumPy, go to your command prompt and type "pip install numpy".
- Once the installation is completed, go to your IDE (For example: Idle/Anaconda)and simply import it by typing: "import numpy as np"
- pip list (will show you all installed packages)

- To check numpy version:

```
import numpy
print(numpy.__version__)
```


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NumPy Arrays v/s List

We use python NumPy array instead of a list because of the below three reasons:

1. Less Memory
2. Fast
3. Convenient

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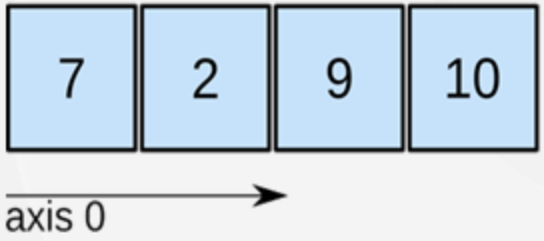
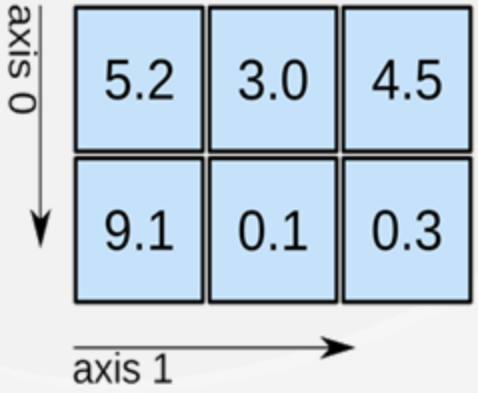
Array

1D array



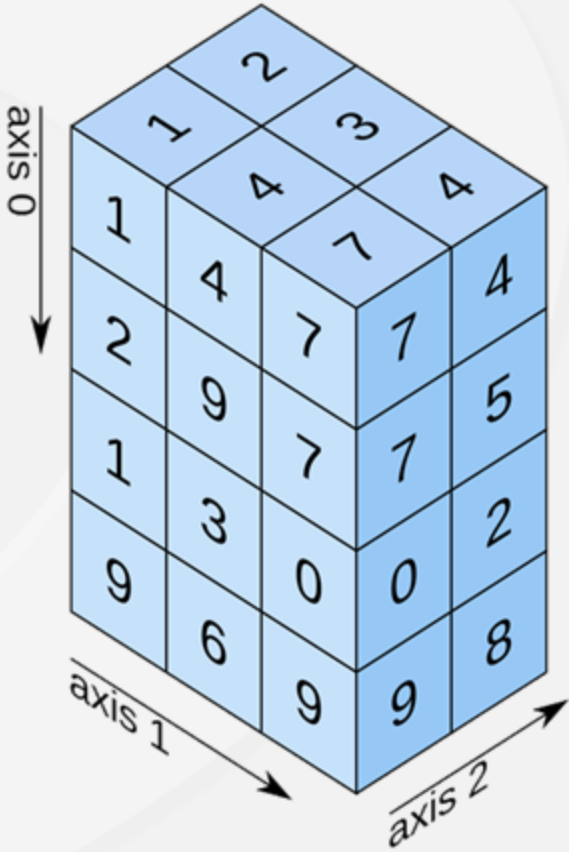
shape: (4,)

2D array



shape: (2, 3)

3D array



shape: (4, 3, 2)

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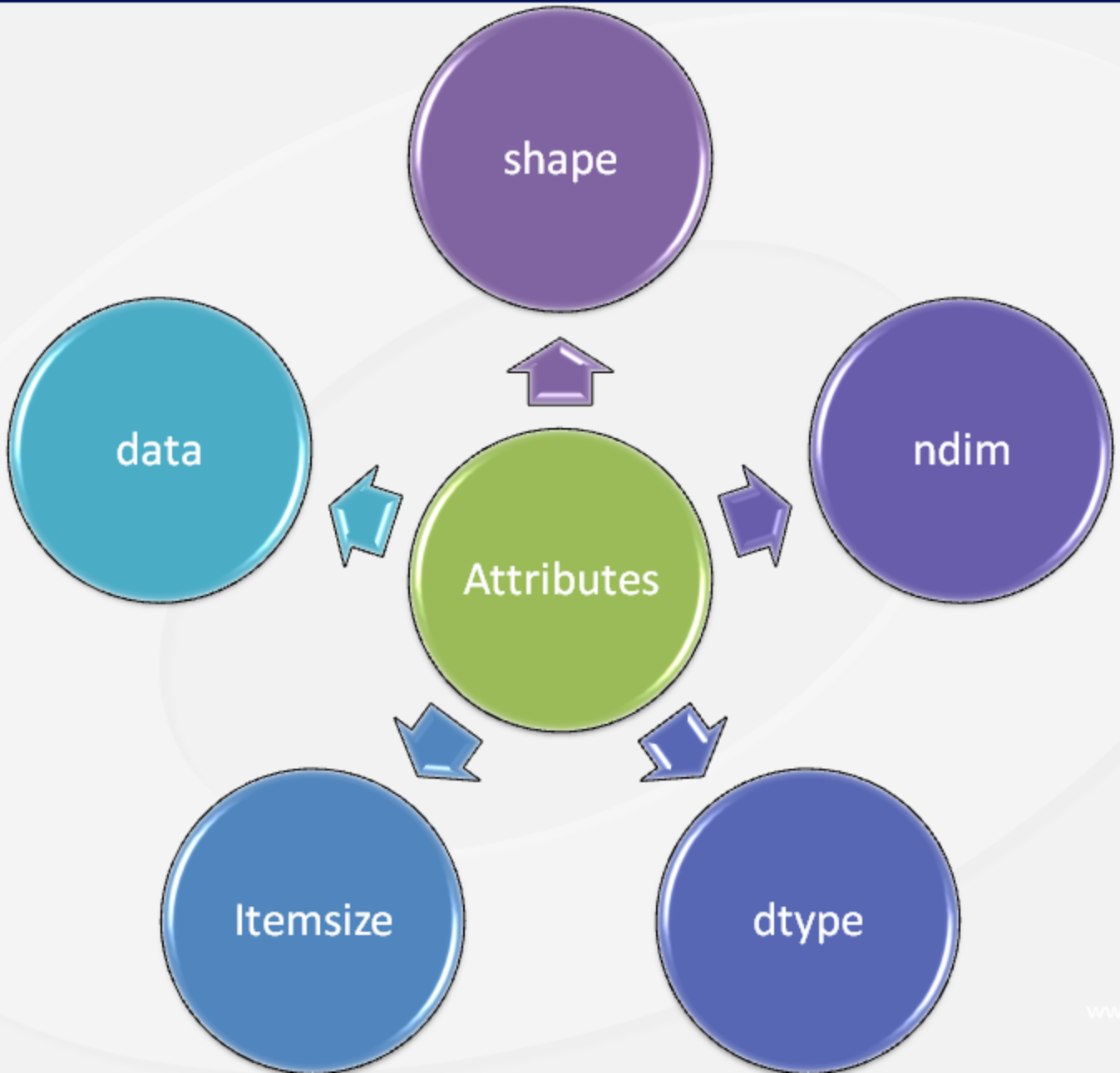
NumPy Attributes



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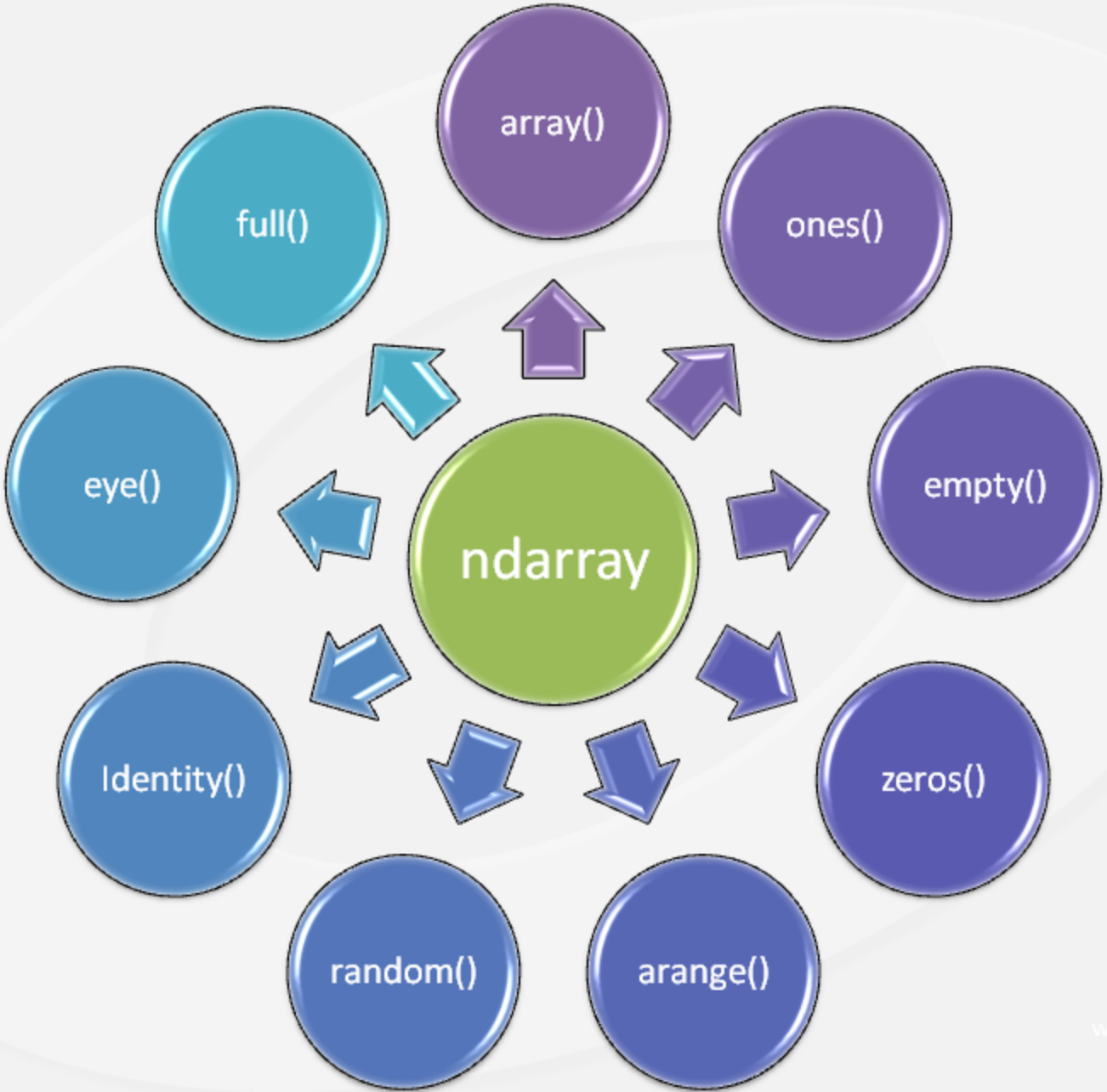
Array Creation



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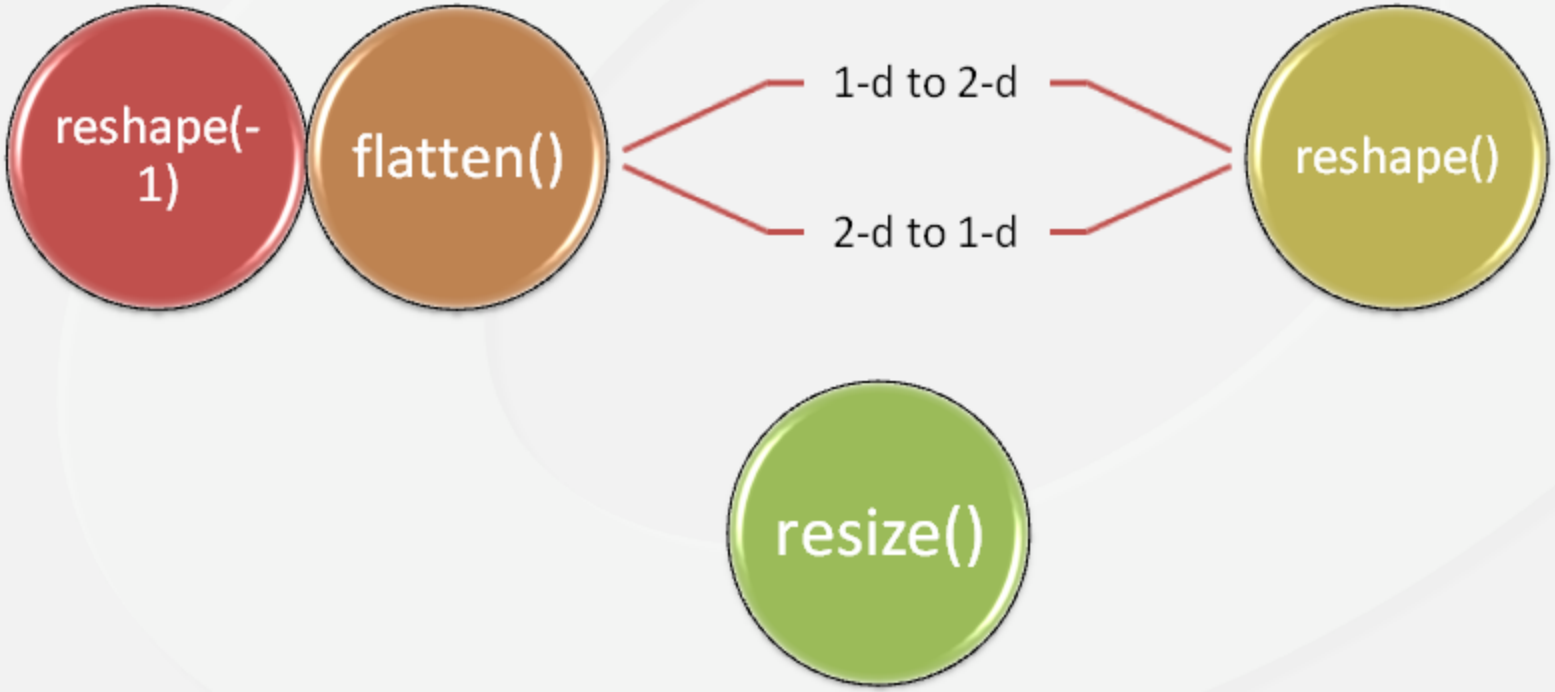
Array Conversion



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Array Indexing/Slicing

```
>>> a[0, 3:5]
array([3, 4])

>>> a[4:, 4:]
array([[44, 55],
       [54, 55]])

>>> a[:, 2]
a([2, 12, 22, 32, 42, 52])

>>> a[2::2, ::2]
array([[20, 22, 24],
       [40, 42, 44]])
```

0	1	2	3	4	5
10	11	12	13	14	15
20	21	22	23	24	25
30	31	32	33	34	35
40	41	42	43	44	45
50	51	52	53	54	55

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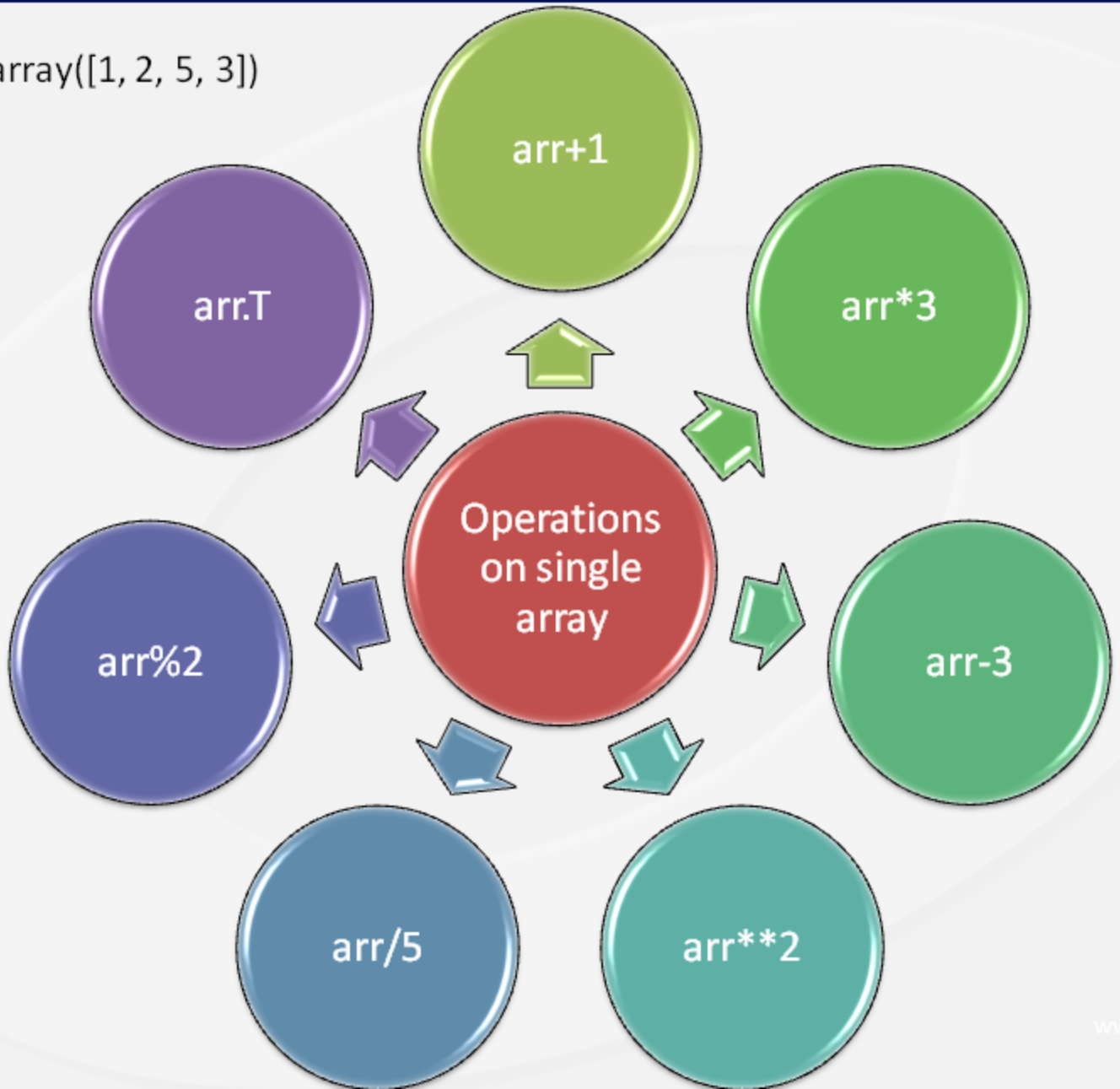
NumPy Operations

arr = np.array([1, 2, 5, 3])



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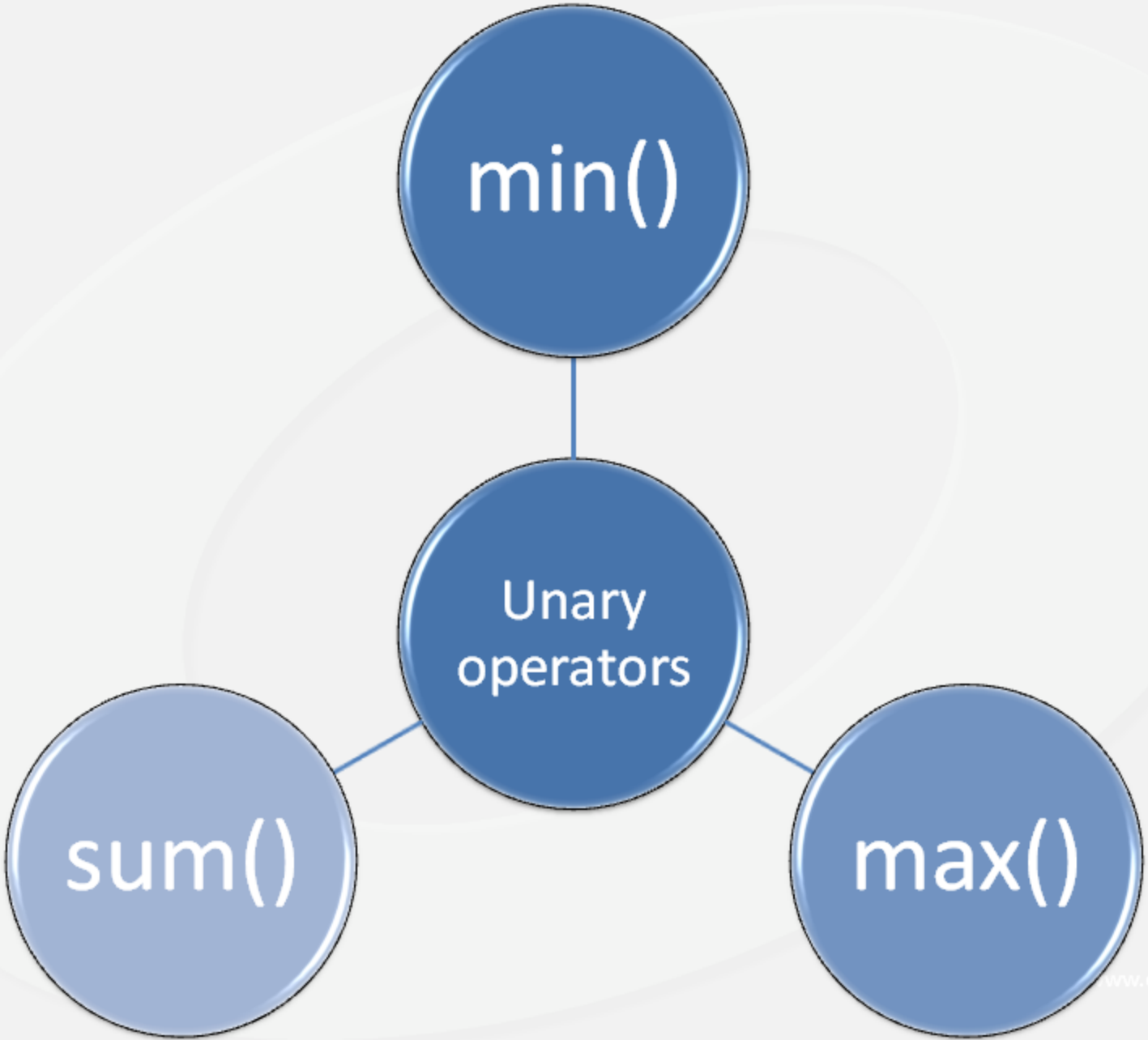
NumPy Unary Operations



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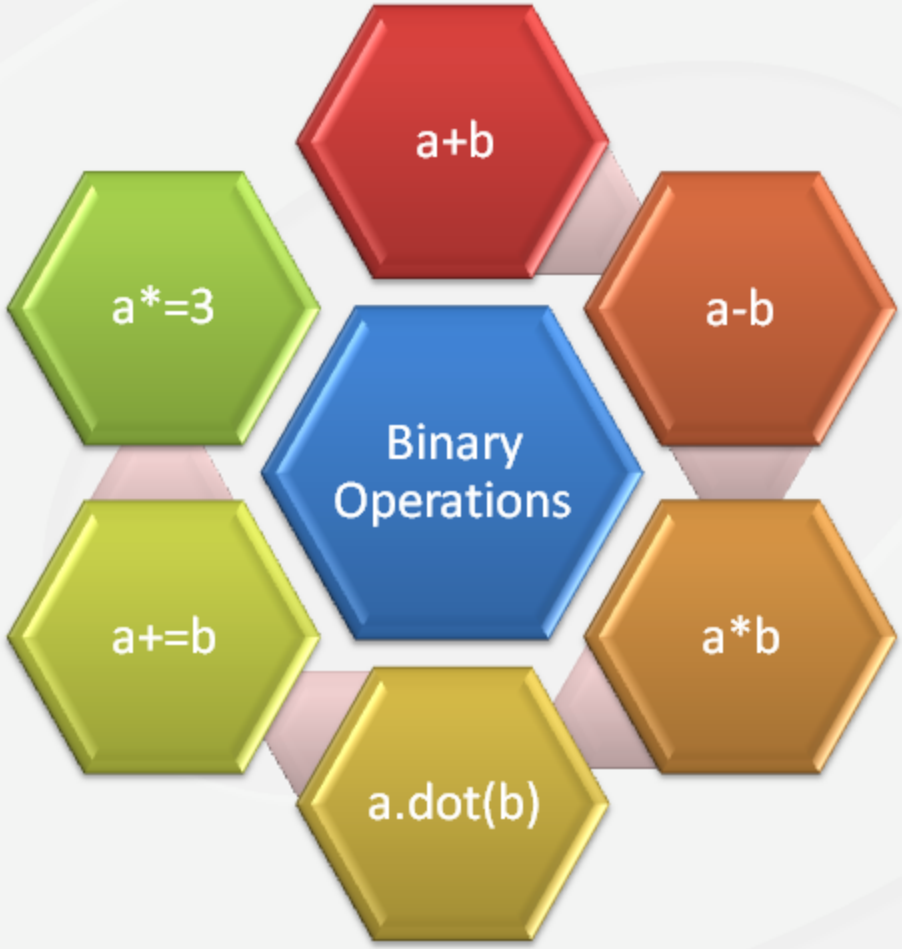
NumPy Binary Operations

```
a = np.array([[1, 2], [3, 4]])
b = np.array([[4, 3], [2, 1]])
```



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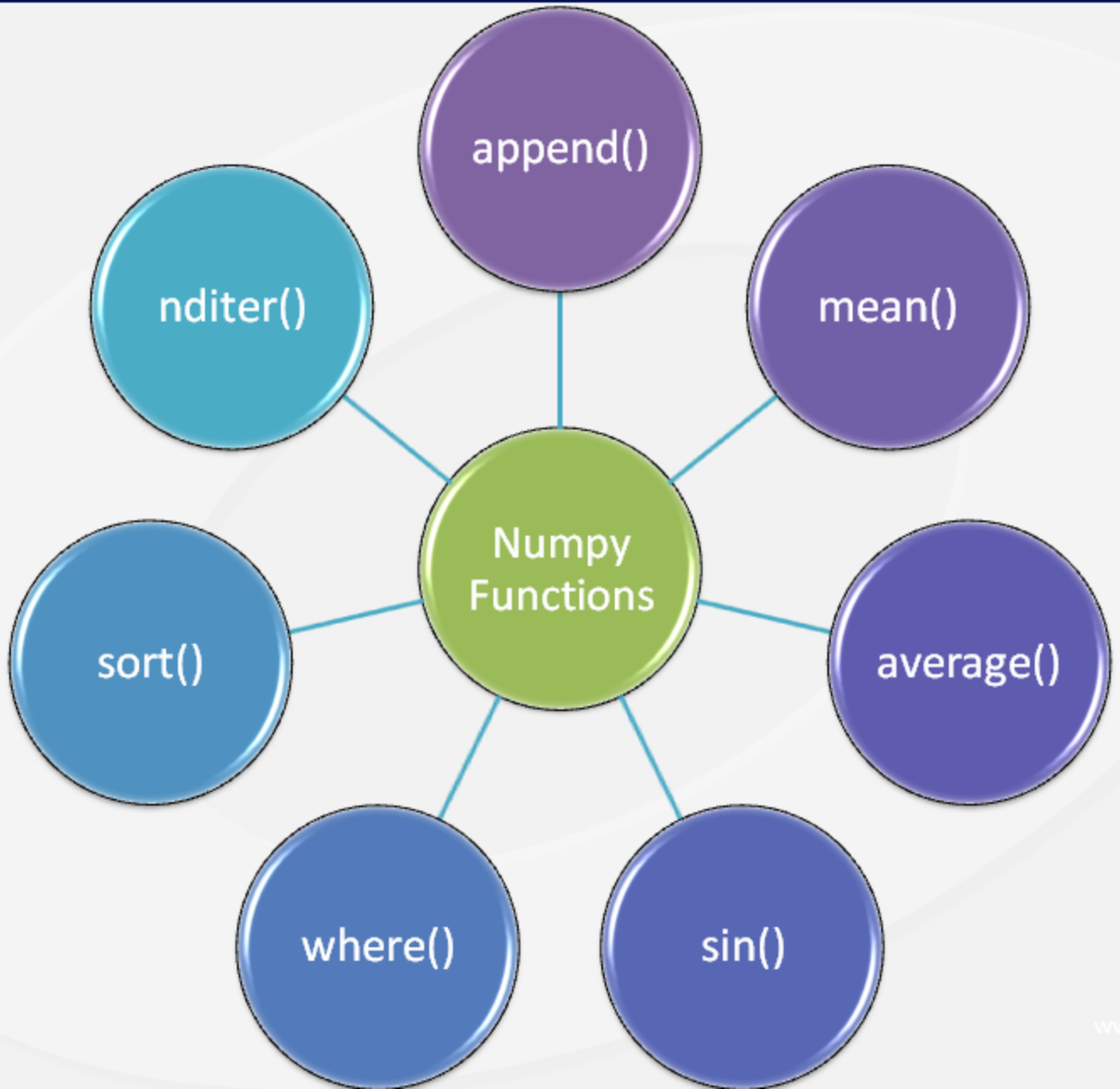
NumPy Special Functions



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Summary

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Thank You